RAG Pipeline

Evaluation Report

Model: **gemma4:e2b** · Generated: 02 May 2026, 10:50 PM

Pipeline Configuration

Configuration	
Retrieval mode	hybrid
Reranker	ON
Adaptive weights	ON
Fusion strategy	RRF
BM25 / Sem weights	0.5 / 0.5 (ADAPTIVE)
Filtering	ON
Distance thresholds	text=1.00 image=0.75, percentile=80
Cross-modal boost	ON
text_k / rerank_k / image_k	10 / 5 / 5
Embed model	BAAI/bge-large-en-v1.5
Image embed model	ViT-L-14 (laion2b_s32b_b82k)
LLM model(s)	gemma4:e2b
Chunk tokens / overlap	384 / 128

Metrics Summary

Retrieval Quality		
M1 Embedding time ↓	12.1468	s
M2 Index size	176 vectors	
M3 Retrieval latency ↓	0.1693	s
■ embed	0.0452	s
■ BM25	0.0031	s
■ fusion	0.0000	s
■ rerank	0.1111	s
M4 Fused Relevance (Text)	0.7462	
M4 Fused Relevance (Image)	0.4190	
M5 Page Coverage@k ↑	89.29	%
M9 Context length	3108	chars

Generation Quality (vs ground truth — 14/14 available)		
M6 ROUGE-1 ↑	0.5108	
M7 ROUGE-2 ↑	0.1925	
M8 ROUGE-L ↑	0.2939	
M10 BLEU ↑	0.1242	
M11 METEOR ↑	0.3468	
M12 BERTScore F1 ↑	0.8771	
M13 FCD ↓	18.29	
M14 Faithfulness ↑	89.86	%
M15 GT Coverage ↑	98.57	%
Performance		
M16 E2E Latency ↓	19.2796	s
M17 Throughput ↑	0.052	q/s
M18 CPU Usage ↓	0.26	%
M19 RAM Usage ↓	0.131	GB
GPU avg	26.36	%
GPU peak	100.00	%

Query 1 of 14

Question

The trajectory diagram of Voyager 1 and Voyager 2 shows clearly diverging paths after Saturn. Using both the trajectory image and the mission text, explain why the two paths diverge at that point, and what specific orbital geometry decision made Voyager 1's post-Saturn path irrecoverable for further planetary encounters.

Retrieved Context

[1] Flight trajectories of Voyager 1 and 2 through the outer solar system (left) and the launch on top of a
Titan III-Centaur rocket (right). Images courtesy of NASA.
Taking advantage of this alignment would be two Voyager spacecraft, both beginning their long journeys in 1977. Voyager 2 launched first, on August 20, followed by Voyager 1 on September 5. Both spacecraft would first fly by Jupiter and use that planet's massive gravity to bend their trajectories to then fly by Saturn. Voyager 1 would also be targeted to fly by Saturn's moon Titan, which was known to have a dense atmosphere, a trajectory that would preclude any future planetary flybys. But the option
(Source: Voyager Grand Tour PDF.pdf, page 1)

[2] Both Voyagers quietly coasted as they headed toward their next target, the ringed planet Saturn. Once again, Voyager 1 was first, beginning observations in August 1980 before making its closest approach 77,000 miles above Saturn's cloud tops on November 12, and finishing studies in December.
Significantly, Voyager 1 made a close flyby of Saturn's large moon Titan, larger than our own Moon and with a dense atmosphere, coming within 4,000 miles of its surface near its South Pole. The Titan encounter changed Voyager 1's trajectory so it could not make any further planetary encounters – thus beginning its long journey out of our solar system. Voyager 2 started its examination of Saturn in June 1981, making its closest approach at 63,000 miles on August 25, and finishing observations in September.
The instruments aboard the Voyagers returned a wealth of new information about Saturn and the cameras returned thousands of images of the planet, its rings and satellites. Among significant discoveries, the Voyagers found that Saturn's rings were far more complex than previously believed, Titan may have lakes of liquid hydrocarbons, Saturn has auroras near its poles much like Earth does, and that there were three previously unknown satellites.
(Source: Voyager Grand Tour PDF.pdf, page 3)

[3] the planet as well as a magnetic field, and geysers on the large moon Triton. After exiting the Neptune system, Voyager 2 joined its fellow spacecraft on a journey out of the solar system. But the intrepid Voyagers weren't quite done yet.
(Source: Voyager Grand Tour PDF.pdf, page 4)

[Image Captions]
[Image 1 Caption] Flight trajectories of Voyager 1 and 2 through the outer solar system (left) and the launch on top of a
Titan III-Centaur rocket (right). Images courtesy of NASA.

Ground Truth Answer

The trajectory diagram shows Voyager 1's path bending sharply away from the ecliptic plane after Saturn while Voyager 2 continues along a shallower arc toward Uranus and Neptune. The text explains this divergence was caused by Voyager 1's close flyby of Titan, coming within 4,000 miles of its surface. That encounter changed Voyager 1's trajectory such that it could not make any further planetary encounters, sending it out of the solar system. Voyager 2 used a different Saturn geometry — a closest approach at 63,000 miles versus Voyager 1's 77,000 miles — that preserved the gravitational slingshot angle needed for the Uranus and Neptune legs. The diagram makes the irreversibility visually concrete: the labeled date nodes on Voyager 2's path continue to Jan 24, 1986 (Uranus) and Aug 25, 1989 (Neptune), while Voyager 1's path terminates its planetary arc at Saturn Nov 12, 1980.

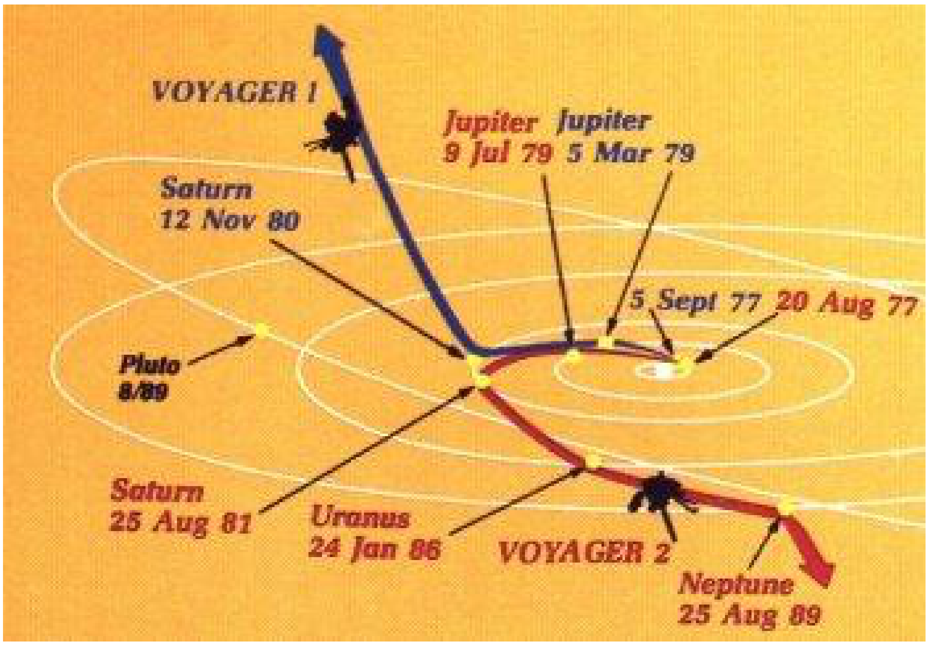
Model Answer

The trajectory diagram shows the flight paths of Voyager 1 and Voyager 2 through the outer solar system. Both spacecraft were initially targeted to fly by Jupiter and then Saturn, using the planet's gravity to bend their trajectories. Voyager 1 was also targeted to fly by Saturn's moon Titan.

The text explains that Voyager 1 made a close flyby of Saturn's large moon Titan, which has a dense atmosphere. This encounter changed Voyager 1's trajectory so that it could not make any further planetary encounters, thus beginning its long journey out of the solar system. This specific encounter made Voyager 1's post-Saturn path irrecoverable for further planetary encounters.

Inference: 28.2941 s · E2E: 28.9637 s · CPU: 0.2% · RAM: 0.438 GB · GPU: 13.6% · Images: 1 · FCD: 13.94

Retrieved Images



Flight trajectories of Voyager 1 and 2 through the outer solar system (left) and the launch on top of a Titan III-Centaur rocket (right). Images courtesy of NASA.

Query 2 of 14

Question

The Earth-Moon photograph taken by Voyager 1 is described as the first single-frame image of both bodies together. Using the image itself and the surrounding mission text, explain what engineering and operational conditions had to be simultaneously satisfied to produce it, and why this milestone was considered a meaningful preview of capabilities rather than merely a publicity photograph.

Retrieved Context

[1] Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million
miles, it turned its camera back toward Earth and snapped the first ever photograph of the Earth-Moon
system in a single frame, giving a sneak preview of the discoveries that lay ahead.
(Source: Voyager Grand Tour PDF.pdf, page 2)

[2] First photograph of the Earth-Moon system in a single frame (left), taken by Voyager 1 on September 18,
1977. Voyager spacecraft as it would appear in space (right). Images courtesy of NASA.
On their way to Jupiter, both Voyagers successfully passed through the Asteroid Belt without incident –
two Pioneer spacecraft had made the crossing safely in 1973 and 1974 as pathfinders, proving that it
was safe to do so. Voyager 1 arrived first, passing 217,000 miles above the giant planet's cloud tops on
March 5, 1979. Two months before the close encounter, Voyager 1 began taking photographs of Jupiter
and its moons and gathering other scientific data about the planet and its environment, a process that
continued for about a month after the fly-by. Voyager 2 followed not too far behind, beginning
scientific observations in April 1979 before passing 350,000 miles above Jupiter's cloud tops on July 9,
1979, and finishing observations about a month later.
The 11 instruments aboard each Voyager made amazing discoveries, not to mention the thousands of
stunning photographs of Jupiter and its many moons. Among the more notable discoveries were that
Jupiter has a faint ring system, its Great Red Spot is a large swirling storm system, the moon Io has active
volcanoes, the moon Europa has a geologically young surface made of ice possibly floating on an ocean
of liquid water, and that there were three previously undiscovered small moons.
(Source: Voyager Grand Tour PDF.pdf, page 2)

[3] was kept open, if Voyager 1's Titan flyby was successful, to retarget Voyager 2 to send it on to Uranus
and maybe even Neptune – assuming it would survive that long!
Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million
miles, it turned its camera back toward Earth and snapped the first ever photograph of the Earth-Moon
system in a single frame, giving a sneak preview of the discoveries that lay ahead.
(Source: Voyager Grand Tour PDF.pdf, page 2)

[Image Captions]
[Image 1 Caption] was kept open, if Voyager 1's Titan flyby was successful, to retarget Voyager 2 to send it on to Uranus
and maybe even Neptune – assuming it would survive that long! Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million
miles, it turned its camera back toward Earth and snapped the first ever photograph of the Earth-Moon
system in a single frame, giving a sneak preview of the discoveries that lay ahead. First photograph of the Earth-Moon system in a single frame (left), taken by Voyager 1 on September 18,
1977. Voyager spacecraft as it would appear in space (right). Images courtesy of NASA.

Ground Truth Answer

The image shows both Earth and the Moon captured together in a single frame from 7.25 million miles, taken just 13 days after launch on September 18, 1977. To produce it, the camera had to achieve stable long-range pointing at two objects separated by significant angular distance, frame both within a single exposure without saturating on

Earth's brightness, and do so while the spacecraft was still in early operational checkout. The text describes it as thefirst of Voyager 1's many firsts, framing it as a sneak preview of the discoveries ahead. The engineering significanceis that it validated the imaging system's ability to handle multi-body framing at interplanetary distances under realflight conditions, not just in laboratory testing. For mission planners, a camera that could jointly frame Earth andMoon from 7.25 million miles had clearly demonstrated the pointing precision and dynamic range that would beneeded to image moons near giant planets from comparable or greater distances during the actual sciencecampaign.

Model Answer

The first photograph of the Earth-Moon system in a single frame was taken by Voyager 1 at a distance of 7.25 million miles, just 13 days after its launch, as part of its first set of discoveries. This achievement required specific engineering and operational conditions to be simultaneously satisfied. The text indicates that this photograph was taken by Voyager 1, which was operating at a distance of 7.25 million miles from Earth, demonstrating the spacecraft's capability to perform this observation.

This milestone was considered a meaningful preview of capabilities because it provided a "sneak preview of the discoveries that lay ahead." This suggests that successfully capturing this image demonstrated the advanced engineering and operational capabilities of the Voyager spacecraft, setting a precedent for future scientific endeavors and hinting at the potential for further discoveries.

Inference: 12.6738 s · E2E: 13.0684 s · CPU: 0.3% · RAM: 0.072 GB · GPU: 17.3% · Images: 1 · FCD: 17.49

Retrieved Images

was kept open, if Voyager 1's Titan flyby was successful, to retarget Voyager 2 to send it on to Uranus and maybe even Neptune – assuming it would survive that long! Just 13 days after its launch, Voyager 1 scored the first of its many firsts: at a distance of 7.25 million miles, it turned its camera back toward Earth and snapped the first ever photograph of the Earth-Moon system in a single frame, giving a sneak preview of the discoveries that lay ahead. First photograph of the Earth-Moon system in a single frame (left), taken by Voyager 1 on September 18, 1977. Voyager spacecraft as it would appear in space (right). Images courtesy of NASA.

Query 3 of 14

Question

Curiosity's Sky Crane landing image and the text description of the landing sequence together tell a more complete story than either alone. Using both the image of the sky crane lowering Curiosity and the technical description of the EDL sequence, explain why the sky crane approach was necessary rather than airbag-based landing used by earlier rovers, and what physical constraints of the crater site made this the only viable solution.

Retrieved Context

[1] ited many design elements from them, including six-wheel
In the first few weeks after landing, images from the rover
showed that Curiosity touched down right in an area
where water once coursed vigorously over the surface.
The evidence for stream flow was in rounded pebbles
mixed with hardened sand in conglomerate rocks at and
near the landing site. Analysis of Mars' atmospheric com-
position early in the mission provided evidence that the
planet has lost much of its original atmosphere by a pro-
cess favoring loss from the top of the atmosphere rather
than interaction with the surface.
In the initial months of the surface mission, the rover team
drove Curiosity eastward toward "Yellowknife Bay" to
Curiosity's first sample drilling, at a rock called "John Klein."
Mars Science Laboratory/Curiosity
2
NASA Facts
(Source: mars-science-laboratory.pdf, page 2)

[2] National Aeronautics and Space Administration
Mars Science Laboratory/Curiosity
NASA's Mars Science Laboratory mission set
down a large, mobile laboratory — the rover
Curiosity — at Gale Crater, using precision
landing technology that made one of Mars'
most intriguing regions a viable destination for
the first time. Within the first eight months of
a 23-month primary mission, Curiosity met its
major objective of finding evidence of a past
environment well suited to supporting microbial
Curiosity touches down on Mars after being lowered by its
Sky Crane.
life. The rover studies the geology and environ-
ment of selected areas in the crater and ana-
lyzes samples drilled from rocks or scooped
The Mars Science Laboratory spacecraft
launched from Cape Canaveral Air Force Sta-
Curiosity carries the most advanced payload
tion, Florida, on Nov. 26, 2011. Mars rover
of scientific gear ever used on Mars' surface,
Curiosity landed successfully on the floor of
a payload more than 10 times as massive as
Gale Crater on Aug. 6, 2012 Universal Time

those of earlier Mars rovers. Its assignment:
(evening of Aug. 5, Pacific Time), at 4.6 degrees
Investigate whether conditions have been
south latitude, 137.4 degrees east longitude
favorable for microbial life and for preserving
and minus 4,501 meters (2.8 miles) elevation.
clues in the rocks about possible past life.
Engineers designed the spacecraft to steer it-
More than 400 scientists from around the
self during descent through Mars' atmosphere
world participate in the science operations.
with a series of S-curve maneuvers similar to
those used by astronauts piloting NASA space
shuttles. During the three minutes before touch-
down, the spacecraft slowed its descent with
a parachute, then used retrorockets mounted
(Source: mars-science-laboratory.pdf, page 1)

[3] capability of landing within a target area only about 20 ki-
lowknife Bay area and began a southwestward trek to
lometers (12 miles) long. That precision, about a five-fold
the base of Mount Sharp. It reached the base layer of
improvement on earlier Mars landings, made sites eligible
this main destination in September 2014. In the low lay-
that would otherwise be excluded for encompassing
ers of Mount Sharp during the rover's extended mission,
nearby unsuitable terrain. The Gale Crater landing site is so
researchers anticipate finding further evidence about hab-
close to the crater wall and Mount Sharp that it would not
itable past environments and about the evolution of the
have been considered safe if the mission were not using
Martian environment from a wetter past to a drier present.
this improved precision.
Science findings began months before landing. Measure-
Curiosity is about twice as long (about 3 meters or 10 feet)
ments that Curiosity made of natural radiation levels during
and five times as heavy as NASA's twin Mars Exploration
the flight from Earth to Mars will help NASA design for as-
Rovers, Spirit and Opportunity, launched in 2003. It inher-
tronaut safety on future human missions to Mars.
ited many design elements from them, including six-wheel
In the first few weeks after landing, images from the rover
showed that Curiosity touched down right in an

[Image Captions]
[Image 1 Caption] Mars Science Laboratory/Curiosity Curiosity touches down on Mars after being lowered by its Sky Crane. The Mars Science Laboratory spacecraft launched from Cape Canaveral Air Force Sta-

Ground Truth Answer

The image shows Curiosity being lowered on a tether from the upper sky crane stage to land wheels-down on the Martian surface. The text explains that Curiosity's payload is more than 10 times as massive as earlier rovers and its science equipment required distributing samples to internal analytical instruments, which necessitated landing upright on wheels rather than rolling in a protective airbag cocoon. The precision landing capability reduced thetarget ellipse to about 20 kilometers, a five-fold improvement over earlier missions. The Gale Crater site was soclose to the crater wall and Mount Sharp that airbag-style landing with the wider uncertainty ellipses of earliermissions would have been unsafe. The sky crane therefore solved two simultaneous problems: it accommodatedCuriosity's size and upright instrument layout while also enabling the precision that unlocked Gale Crater as ascientifically viable destination.

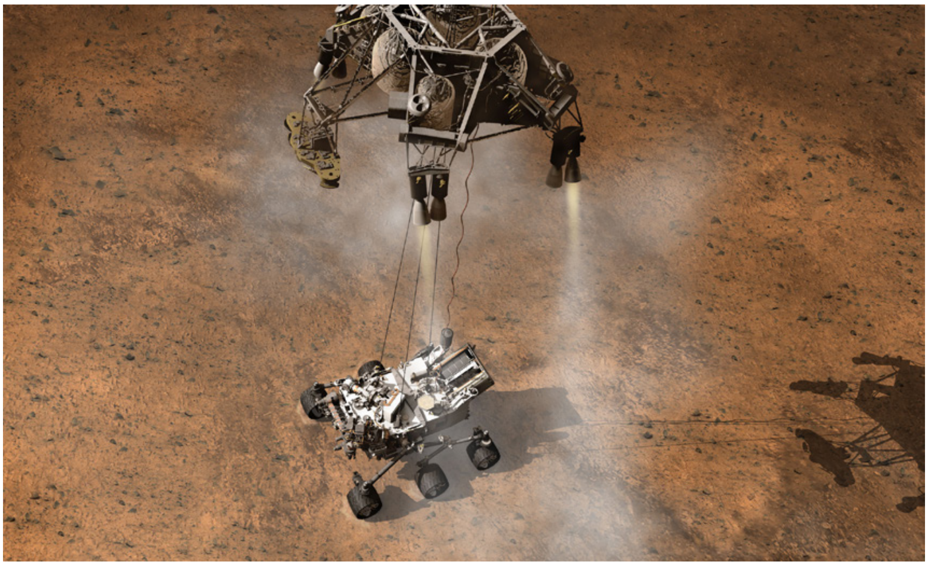
Model Answer

The image of the Sky Crane lowering Curiosity and the text description of the Entry, Descent, and Landing (EDL) sequence together provide a more complete story regarding the landing process. The text states that Curiosity touched down on Mars after being lowered by its Sky Crane, which used a series of S-curve maneuvers similar to those used by astronauts piloting NASA space shuttles during descent through Mars' atmosphere. This process involved slowing the descent with a parachute and then using retrorockets mounted on the spacecraft during the three minutes before touchdown.

This method was necessary because the mission required precision landing within a target area, which the text notes was achieved with "precision, about a five-fold the base of Mount Sharp." The text implies that this precision was crucial for making the Gale Crater landing site a viable destination for the first time. The context suggests that the precision afforded by the Sky Crane approach was essential because the landing site was one of Mars' most intriguing regions, and the engineers designed the spacecraft to steer it with this capability. The text does not explicitly state that airbag-based landing was unsuitable, but it emphasizes the need for precise navigation and control during descent, which the Sky Crane system provided to achieve the required landing accuracy.

Inference: 22.7002 s · E2E: 23.2589 s · CPU: 0.3% · RAM: 0.088 GB · GPU: 31.1% · Images: 1 · FCD: 29.53

Retrieved Images



Mars Science Laboratory/Curiosity Curiosity touches down on Mars after being lowered by its Sky Crane. The Mars Science Laboratory spacecraft launched from Cape Canaveral Air Force Sta

Query 4 of 14

Question

The image of the rock outcrop called Link and the image of the first drill sample at John Klein represent two distinct evidentiary steps in Curiosity's habitability argument. Using both images and the corresponding text, explain why the visual evidence from Link was scientifically necessary but not sufficient, and what the John Klein drilling contributed that images alone could never provide.

Retrieved Context

[1] investigate an ancient river and fan system identified in
The rover analyzed its first scoops of soil on the way to
Yellowknife Bay. Once there, it collected the first samples
of material ever drilled from rocks on Mars. Analysis of the
first drilled sample, from a rock target called “John Klein,”
provided the evidence of conditions favorable for life in
Mars’ early history: geological and mineralogical evidence
for sustained liquid water, other key elemental ingredients
for life, a chemical energy source, and water not too acid-
ic or too salty. On a subsequent drill sample, Curiosity
A rock outcrop called Link shows signs of being formed by the deposition
of water.
was able to accomplish a first for measurements on
another planet: determining the age of the rock. The
lowed consideration of more than 30 Martian locations by
measurements showed that the drilled material was
more than 100 scientists participating in a series of open
4.2 billion years old and yet had been exposed at the
workshops. The selection process benefited from examin-
surface for only 80 million years.
ing candidate sites with NASA’s Mars Reconnaissance
Orbiter and earlier orbiters, and from the rover mission’s
In July 2013, Curiosity finished investigations in the Yel-
capability of landing within a target area only about 20 ki-
lowknife Bay area and began a southwestward trek to
lometers (12 miles) long. That precision, about a five-fold
the base of Mount Sharp. It reached the base layer of
improvement on earlier Mars landings, made sites eligible
this main destination in September 2014. In the low lay-
that would otherwise be excluded for encompassing
ers of Mount Sharp during the rover’s extended mission,
nearby unsuitable terrain. The Gale Crater landing site is so
researchers anticipate finding further evidence about hab-
(Source: mars-science-laboratory.pdf, page 2)

[2] ited many design elements from them, including six-wheel
In the first few weeks after landing, images from the rover
showed that Curiosity touched down right in an area
where water once coursed vigorously over the surface.
The evidence for stream flow was in rounded pebbles
mixed with hardened sand in conglomerate rocks at and
near the landing site. Analysis of Mars’ atmospheric com-
position early in the mission provided evidence that the

planet has lost much of its original atmosphere by a pro-
cess favoring loss from the top of the atmosphere rather
than interaction with the surface.
In the initial months of the surface mission, the rover team
drove Curiosity eastward toward “Yellowknife Bay” to
Curiosity’s first sample drilling, at a rock called “John Klein.”
Mars Science Laboratory/Curiosity
2
NASA Facts
(Source: mars-science-laboratory.pdf, page 2)

[3] samples of rocks and soil, distributing them to onboard
atmosphere and water. The principal investigator is Paul
test chambers inside analytical instruments.
Mahaffy of NASA’s Goddard Space Flight Center, Green-
NASA’s Jet Propulsion Laboratory (JPL), Pasadena, Calif.,
builder of the Mars Science Laboratory, engineered Curi-
An X-ray diffraction and fluorescence instrument called
osity to roll over obstacles up to 65 centimeters (25 inch-
CheMin also examines samples gathered by the robotic
es) high and to travel up to about 200 meters (660 feet)
arm. It is designed to identify and quantify the minerals in
per day on Martian terrain.
rocks and soils, and to measure bulk composition. The
principal investigator is David Blake of NASA’s Ames Re-
The rover’s electrical power is supplied by a U.S. Depart-
search Center, Moffett Field, Calif.
ment of Energy radioisotope power generator. The multi-
mission radioisotope thermoelectric generator produces
Mounted on the arm, the Mars Hand Lens Imager takes
electricity from the heat of plutonium-238’s radioactive
extreme close-up pictures of rocks and soil, revealing
decay. This long-lived power supply gives the mission an
details smaller than the width of a human hair. It can also
operating lifespan on Mars’ surface of a full Mars year
focus on hard-to-reach objects more th

[Image Captions]
[Image 1 Caption] A rock outcrop called Link shows signs of being formed by the deposition
of water. lowed consideration of more than 30 Martian locations by more than 100 scientists participating in a series of open workshops. The selection process benefited from examin-

Ground Truth Answer

The Link outcrop image shows rounded pebbles mixed with hardened sand in conglomerate rock, which the text identifies as evidence for past vigorous water flow — the rounding and sorting of the pebbles indicates sustained fluvial transport. However, visual evidence of ancient stream flow establishes only water presence and movement, not water chemistry, persistence, or geochemical habitability. The John Klein drill image shows the first borehole ever made in Martian rock, and the text explains that analysis of that drilled sample provided the full habitability evidence package: geological and mineralogical evidence for sustained liquid water, key elemental ingredients for life, a chemical energy source, and water that was neither too acidic nor too salty. Images could establish morphology and sedimentary context; only in situ chemical analysis of drilled interior material could constrain whether the ancient environment was genuinely suited to microbial life.

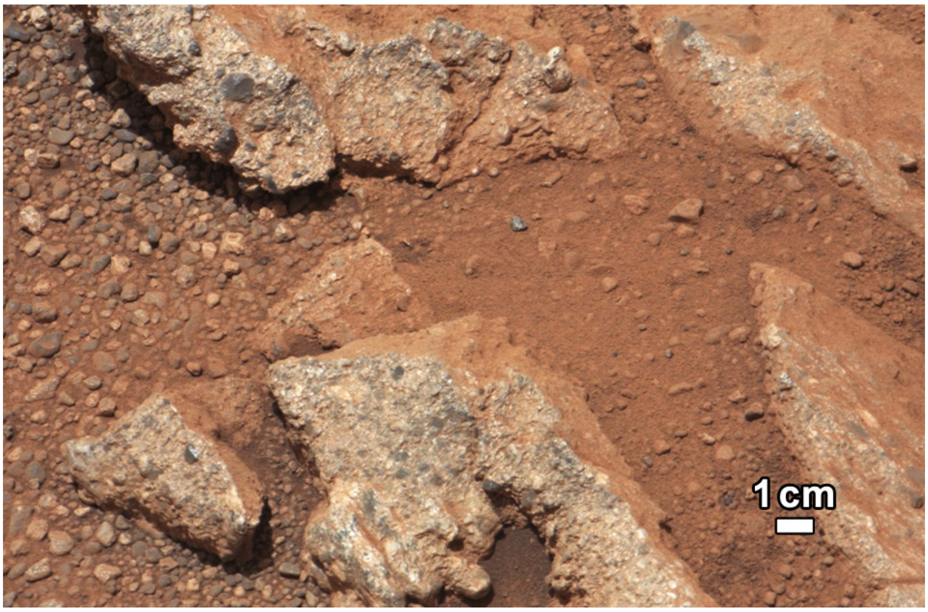
Model Answer

The image of the rock outcrop called Link shows signs of being formed by the deposition of water, which provides geological evidence related to past water activity. This visual evidence is scientifically necessary because it indicates a history of water interaction on Mars. However, the text indicates that this evidence alone is not sufficient.

The drilling of the first sample at the rock target called "John Klein" contributed crucial, quantitative evidence that images alone could never provide. Analysis of this drilled sample provided evidence of conditions favorable for life in Mars' early history, including geological and mineralogical evidence for sustained liquid water, other key elemental ingredients for life, a chemical energy source, and water that was not too acidic or too salty. Therefore, while the outcrop at Link suggests past water flow, the drilling at John Klein provided the necessary chemical and mineralogical data to support the habitability argument.

Inference: 13.7286 s · E2E: 14.3510 s · CPU: 0.3% · RAM: 0.096 GB · GPU: 25.1% · Images: 1 · FCD: 16.63

Retrieved Images



A rock outcrop called Link shows signs of being formed by the deposition of water. lowed consideration of more than 30 Martian locations by more than 100 scientists participating in a series of open workshops. The selection process benefited from examin

Query 5 of 14

Question

The Hubble deep field image and the panel of faint early galaxies shown alongside it are described as evidence for galaxy growth over cosmic time. Using the visual appearance of those galaxies and the text about hierarchical galaxy formation, explain how the morphological evidence in the images connects to the theoretical prediction that large present-day galaxies assembled through mergers rather than forming fully intact.

Retrieved Context

[1] Tracing the Growth of Galaxies
Like documenting a child's development in a scrapbook,
astronomers have used Hubble to capture the appearance
of many developing galaxies throughout cosmic time. This is
possible because of the mathematical relationship between
cosmic distance and time: the deeper Hubble peers into
space, the farther back it looks in time. As it happens, the
most distant and earliest galaxies spied by Hubble are smaller
and more irregularly shaped than today's grand spiral and
elliptical galaxies. This is evidence that galaxies grew over
time through mergers with other galaxies to become the giant
systems we see today.
amounts of stars and gas in galaxies, the types and amounts of
identifiable chemical elements present, and star-formation rates.
And the evolution continues. Hubble observations of
our neighboring Andromeda galaxy (M31) have allowed
astronomers to predict with certainty that a titanic collision
between our Milky Way galaxy and Andromeda will inevitably
take place beginning 4 billion years from now. The galaxy is
now 2.5 million light-years away, but it is inescapably moving
toward the Milky Way under the mutual pull of gravity between
the two galaxies and the invisible dark matter that surrounds
them both. The merger will likely result in the creation of a
giant elliptical galaxy billions of years from now.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 4)

[2] Because the universe was smaller in the past, galaxies were
more likely to interact with one another gravitationally. Some of
Hubble's cosmic "snapshots" show fantastic stellar streamers
pulled out and flung across space by colliding galaxies. They
apparently settled over time into the more familiarly shaped
galaxies seen closer to Earth and hence nearer to the present
time. By carefully studying galaxies at different epochs,
astronomers can see how galaxies changed and evolved
over time. Among the things they investigate are the relative
Left: Hubble's Ultra Deep Field is one of the most distant looks into
space ever. The cumulative exposure time needed to capture the
image was about a million seconds (11 days).
Right: A sample of the faintest and farthest galaxies in the Hubble
Ultra Deep Field shows that they were irregularly shaped and
frequently interacting in the distant past.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 4)

[3] Viewing Galactic Details and Mergers
When Edwin Hubble discovered that the universe was a vast frontier of innumerable galaxies beyond our Milky Way, he categorized them according to three basic shapes: spiral, elliptical, and irregular. Later, the sharp vision of the space telescope named in his honor revealed unprecedented details in such galaxies. In addition, Hubble's images uncovered a plethora of oddball, peculiarly shaped galaxies that are more numerous the farther back into time the telescope looks. This is because the expanding universe was smaller long ago, and galaxies were both younger and more likely to interact since they were closer to one another.
Among this zoo of odd galaxies are "tadpole-like'' objects and apparently merging systems dubbed "train-wrecks." For all their violence, galactic collisions take place at a glacial rate by human standards—timescales on the order of several hundred million years. Hubble's images, therefore, capture snapshots of galaxies at various stages of interaction. The merging of galaxies produces turbulence and tides that can induce new vigorous bursts of star formation within their interstellar gas clouds. These observed mergers form a preview of the coming collision between our own Milky Way and the neighboring Andromeda galaxy 4 billion years from now.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 18)

[Image Captions]
[Image 1 Caption] Left: Hubble's Ultra Deep Field is one of the most distant looks into space ever. The cumulative exposure time needed to capture the image was about a million seconds (11 days). Right: A sample of the faintest and farthest galaxies in the Hubble Ultra Deep Field shows that they were irregularly shaped and frequently interacting in the distant past.

Ground Truth Answer

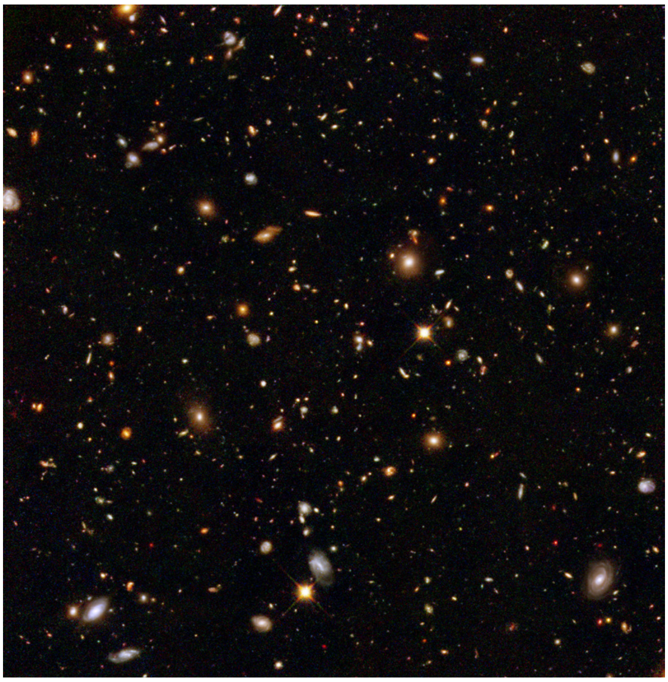
The Ultra Deep Field image reveals thousands of galaxies at varying distances, and the close-up panel of faint, distant galaxies shows they are small, irregular, and frequently interacting — very different from the grand spiral and elliptical galaxies seen in the nearby universe. The text states that the most distant and earliest galaxies are smaller and more irregularly shaped, and that the universe was smaller in the past, making gravitational interactions between galaxies more frequent. The visual evidence of disturbed morphologies, asymmetric shapes, and apparent mergers in the distant panels directly supports hierarchical growth: these compact, irregular systems are the raw building blocks that collide and merge over billions of years into the familiar large galaxies seen nearby. The deep field effectively provides a time-ordered fossil record where increasing distance corresponds to increasing lookback time, allowing morphological evolution to be read directly from the images.

Model Answer

The Hubble deep field image and the sample of faint, early galaxies shown alongside it provide visual evidence supporting the theory of galaxy growth over cosmic time through mergers. The text states that the most distant and earliest galaxies spied by Hubble are smaller and more irregularly shaped than today's grand spiral and elliptical galaxies. This morphological evidence indicates that galaxies grew over time through mergers with other galaxies to become the giant systems observed today. Furthermore, the text notes that some of Hubble's cosmic "snapshots" show stellar streamers pulled out and flung across space by colliding galaxies, which apparently settled over time into the more familiarly shaped galaxies seen closer to Earth and nearer to the present time. This observation, combined with the fact that the universe was smaller in the past, making galaxies more likely to interact gravitationally, suggests that the irregular shapes and interactions seen in the distant past are precursors to the more organized structures of the present. The images capture snapshots of galaxies at various stages of interaction,showing them in states of merging, which aligns with the prediction that future large galaxies, such as the eventualcollision between the Milky Way and Andromeda, will result in the creation of a giant elliptical galaxy.

Inference: 20.5944 s · E2E: 21.1738 s · CPU: 0.3% · RAM: 0.104 GB · GPU: 30.5% · Images: 1 · FCD: 16.88

Retrieved Images



Left: Hubble's Ultra Deep Field is one of the most distant looks into space ever. The cumulative exposure time needed to capture the image was about a million seconds (11 days). Right: A sample of the faintest and farthest galaxies in the Hubble Ultra Deep Field shows that they were irregularly shaped and frequently interacting in the distant past.

Query 6 of 14

Question

The dark matter page shows two views of galaxy cluster Cl 0024+17 — one in visible light showing blue arcs, and one with a blue dark matter density overlay. Using both images together with the gravitational lensing text, explain the full chain of inference that connects the visible arc morphology in the first image to the quantitative dark matter mass estimate cited in the text.

Retrieved Context

[1] Top left: This sketch shows paths of light from a distant galaxy that is being gravitationally lensed. Top right: Gravitational lensing has created
three distorted images of the same background galaxy and five of a background quasar in this image of galaxy cluster SDSS J1004+4112.
Bottom: These two pictures show a massive galaxy cluster, Cl 0024+17. The visible-light image (left) shows blue arcs among the yellowish
galaxies. These arcs are magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue
overlay (right) shows the dark matter density needed to account for the gravitational distortions.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 6)

[2] Shown here are two Hubble views of the galaxy M84. Left: This camera view shows the bright core at the center of the galaxy surrounded by
a (vertical) dark band of gas and dust. Right: This plot was generated by passing light near the core of the galaxy through a spectrograph. The
thin, vertical rectangle in the center of the left panel shows the size and shape of the spectrograph's sampling slit. Spectra taken left and right
of the slit's position, which is centered on the core of the galaxy, show a dramatic shift in color to blue or red. Blueshifted light indicates that the
emitting light source is moving toward Earth, while redshifted light indicates objects that are moving away. At 880,000 miles per hour, stars and
glowing gases nearest to the core of M84 are moving the fastest. They are circling a black hole at the center of the galaxy, and so appear to be
moving rapidly toward Earth on the left and receding on the right.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 7)

[3] Shining a Light on Dark Matter
Dark matter is an invisible form of matter that makes up most of the universe's mass and creates its underlying structure. Dark
matter's gravity drives normal matter (gas and dust) to collect and build up into stars and galaxies. Although astronomers cannot
see dark matter, they can detect its influence by observing how the gravity of massive galaxy clusters, which contain dark matter,
bends and distorts the light of more distant galaxies located behind the cluster. This phenomenon is called gravitational lensing.
Hubble's uniquely sharp vision allows astronomers to map the distribution of dark matter in space using gravitational lensing.
Large galaxy clusters contain both dark matter and normal matter. By observing the areas around massive clusters of galaxies
astronomers can identify warped background galaxies and reverse-engineer their distortions to reveal where the densest
concentrations of matter lie. Mathematical models of these results shed light on the location and properties of the lensing

material, both visible and invisible (dark). The universe appears to have about five times more dark matter than regular matter
and seems to be organized around an immense network of dark matter filaments that have grown over time. Massive visible
structures, like galaxy clusters, are found at the intersections of these filaments.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 6)

[Image Captions]
[Image 1 Caption] Top left: This sketch shows paths of light from a distant galaxy that is being gravitationally lensed. Top right: Gravitational lensing has created
three distorted images of the same background galaxy and five of a background quasar in this image of galaxy cluster SDSS J1004+4112. Bottom: These two pictures show a massive galaxy cluster, Cl 0024+17. The visible-light image (left) shows blue arcs among the yellowish
galaxies. These arcs are magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue
overlay (right) shows the dark matter density needed to account for the gravitational distortions.

Ground Truth Answer

The visible-light image of Cl 0024+17 shows blue arc-shaped features among the yellowish cluster galaxies. The text explains these are magnified, distorted images of background galaxies whose light has been bent by the cluster's gravity — a phenomenon called gravitational lensing. The second image overlays in blue the dark matter density distribution required to mathematically account for the observed distortions. The inference chain is: (1) measure the shapes, positions, and degree of distortion of the background arcs; (2) mathematically reverse-engineer the mass distribution needed to produce those specific distortions; (3) compare that total mass to the luminous matter visible in the cluster. The text states the result is that the universe appears to contain about five times more dark matter than regular matter, and that large structures are found at intersections of immense dark matter filament networks. The two images together show both the raw observational input (arcs) and the derived mass product (blue overlay), making the lensing inference chain visually complete.

Model Answer

The text explains that gravitational lensing occurs because the gravity of massive galaxy clusters, which contain dark matter, bends and distorts the light of more distant galaxies located behind the cluster. This phenomenon is called gravitational lensing, and Hubble's vision allows astronomers to map the distribution of dark matter in space using this effect.

The images of galaxy cluster Cl 0024+17 provide a visual representation of this process. The visible-light image on the left shows blue arcs among the yellowish galaxies. These arcs are described as magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue overlay on the right shows the dark matter density required to account for these gravitational distortions.

Therefore, the visible arc morphology in the first image is the distorted light of background galaxies caused by the bending of light due to the gravitational influence of the dark matter within the cluster. By observing these warped galaxies and reverse-engineering their distortions, astronomers can reveal the location and properties of the lensing material, which includes both visible and invisible dark matter, leading to mathematical models that shed light on the location and properties of the lensing material.

Inference: 19.1911 s · E2E: 19.6122 s · CPU: 0.2% · RAM: 0.105 GB · GPU: 28.2% · Images: 1 · FCD: 17.59

Retrieved Images

Top left: This sketch shows paths of light from a distant galaxy that is being gravitationally lensed. Top right: Gravitational lensing has created three distorted images of the same background galaxy and five of a background quasar in this image of galaxy cluster SDSS J1004+4112. Bottom: These two pictures show a massive galaxy cluster, Cl 0024+17. The visible-light image (left) shows blue arcs among the yellowish galaxies. These arcs are magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue overlay (right) shows the dark matter density needed to account for the gravitational distortions.

Query 7 of 14

Question

The image series of Supernova 1987A from 1994 through 2006 shows a ring of material progressively lighting up over twelve years. Using both the time-sequence images and the text description of the shock wave dynamics, explain what physical process the brightening ring documents, why it is observable only over a multi-year baseline, and what this reveals about the pre-explosion mass loss history of the progenitor star.

Retrieved Context

[1] Sept. 24, 1994
Feb. 6, 1998
Mar. 23, 2001
Supernova 1987A as seen by
Hubble in 1994.
Jan. 5, 2003
Dec. 15, 2004
Dec. 6, 2006
A shock wave of material from the central, exploded star of SN 1987A is slamming into an
existing ring of material around the doomed star. As it does, various spots in the ring are heated
and begin to glow. The nature of the strangely shaped pink feature in the center of the ring is
not well understood.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 15)

[2] A shock wave of material from the central, exploded star of SN 1987A is slamming into an
existing ring of material around the doomed star. As it does, various spots in the ring are heated
and begin to glow. The nature of the strangely shaped pink feature in the center of the ring is
not well understood.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 15)

[Image Captions]
[Image 1 Caption] Supernova 1987A as seen by
Hubble in 1994.

Ground Truth Answer

The six-panel image sequence shows SN 1987A's middle ring going from a faint structure in 1994 to a brightening necklace of hot spots by 2006. The text explains that an expanding shock wave from the central explosion is slamming into a pre-existing ring of material surrounding the dead star. As the blast wave reaches and compresses ring material, those sections are heated and begin to glow — the sequentially brightening spots represent the shock front progressively encountering denser regions around the ring's circumference. The process requires a multi-year baseline because the shock wave must physically travel from the explosion center to the ring, and the ring's finite circumference means different azimuthal sections are reached at slightly different times. The existence of the ring itself implies the progenitor star shed a structured shell of material in an asymmetric mass-loss event before it exploded, providing direct evidence that massive stars undergo significant pre-supernova mass ejection that shapes the circumstellar environment the eventual blast wave must propagate through.

Model Answer

The provided text describes that a shock wave of material from the central, exploded star of Supernova 1987A is slamming into an existing ring of material around the doomed star. As this interaction occurs, various spots in the ring are heated and begin to glow. The nature of a strangely shaped pink feature in the center of the ring is not well understood.

This process documents the heating of various spots in the ring as the shock wave interacts with the surrounding material. The brightening ring is observable over a multi-year baseline because the interaction between the shock wave and the ring is a dynamic process that unfolds over time. The text explicitly states that the shock wave is " slamming into" the ring, causing the spots to heat and glow. This observation reveals information about the pre-explosion mass loss history of the progenitor star through the dynamics of this interaction.

Inference: 12.9266 s · E2E: 13.3461 s · CPU: 0.3% · RAM: 0.107 GB · GPU: 23.3% · Images: 1 · FCD: 13.68

Retrieved Images

Supernova 1987A as seen by Hubble in 1994.

Query 8 of 14

Question

The Saturn montage image shows the planet alongside several of its moons at different scales. Using the image to identify visible moons and the text description of Voyager 1's Titan encounter, explain the specific trade-off that made Titan the highest-priority Saturn target, why that priority decision permanently closed Voyager 1's path to the outer planets, and what Voyager 2's different Saturn geometry preserved for the mission as a whole.

Retrieved Context

[1] Montage of Jupiter and its four largest moons, (left) and of Saturn and several of its moons (right).
Images are not to scale. Courtesy of NASA.
Two more stops: Uranus and Neptune
Voyager 2's flyby of Saturn set it up to automatically fly by Uranus after a coast of five and a half years.
It began to study the Uranian system in November 1985 and flew within 50,600 miles of its cloud tops
on January 24, 1986, before concluding its observations in February. In conducting the first ever robotic
reconnaissance of Uranus, Voyager 2 observed the planet's atmosphere, satellites and thin ring system,
discovering 11 new moons and a previously unknown magnetic field and returning about 8,000 pictures.
And then, it was on to Neptune, arriving there after another coast of three and a half years. Beginning
its study in June 1989, Voyager 2 flew within just 3,076 miles to Neptune's cloud tops on August 25,
concluding its observations in October. That close pass to Neptune allowed detailed observations of its
largest moon Triton. During this first ever close up study of Neptune, Voyager 2 returned about 10,000
photographs of the planet, its satellites and dark rings, discovering 6 new moons, a Great Dark Spot on
(Source: Voyager Grand Tour PDF.pdf, page 3)

[2] Montage of Uranus and several of its larger moons (left) and of Neptune and its largest moon Triton
(right). Courtesy of NASA.
On February 14, 1990, more than 12 years after it began its journey and shortly before its cameras were
permanently turned off to conserve power, Voyager 1 spun around and pointed them back into the
solar system. In a mosaic of 60 images, it was able to capture a "family portrait" of six of the solar
system's planets, including a pale blue dot called Earth. It was fitting that these were the last pictures
from either Voyager.
(Source: Voyager Grand Tour PDF.pdf, page 4)

[3] Both Voyagers quietly coasted as they headed toward their next target, the ringed planet Saturn. Once
again, Voyager 1 was first, beginning observations in August 1980 before making its closest approach
77,000 miles above Saturn's cloud tops on November 12, and finishing studies in December.
Significantly, Voyager 1 made a close flyby of Saturn's large moon Titan, larger than our own Moon and
with a dense atmosphere, coming within 4,000 miles of its surface near its South Pole. The Titan
encounter changed Voyager 1's trajectory so it could not make any further planetary encounters – thus
beginning its long journey out of our solar system. Voyager 2 started its examination of Saturn in June
1981, making its closest approach at 63,000 miles on August 25, and finishing observations in
September.
The instruments aboard the Voyagers returned a wealth of new information about Saturn and the
cameras returned thousands of images of the planet, its rings and satellites. Among significant
discoveries, the Voyagers found that Saturn's rings were far more complex than previously believed,
Titan may have lakes of liquid hydrocarbons, Saturn has auroras near its poles much like Earth does, and
that there were three previously unknown satellites.
(Source: Voyager Grand Tour PDF.pdf, page 3)

[Image Captions]
[Image 1 Caption] Family portrait of six planets taken by Voyager 1 on February 14, 1990, when it was 6 B km from
Earth. Not exactly close encounters

Ground Truth Answer

The Saturn montage shows the planet's rings and several moons, with Titan visible as a large body. The text states Titan is larger than Earth's own Moon and was known to have a dense atmosphere, making it uniquely scientifically valuable among Saturn's satellites. Voyager 1 flew within 4,000 miles of Titan's surface near its South Pole — the text explicitly states this Titan encounter changed Voyager 1's trajectory so it could not make any further planetary encounters. Voyager 2's closest Saturn approach was at 63,000 miles, a different geometry that did not incur the same trajectory penalty. The consequence was that Voyager 2's Saturn flyby automatically set up a coast of five and a half years to Uranus, where it discovered 11 new moons, and then continued to Neptune where it discovered 6 more moons and observed geysers on Triton. The two-spacecraft strategy deliberately split the objectives: one vehicle maximized Titan science at the cost of the extended tour, while the other preserved the grand-tour continuation.

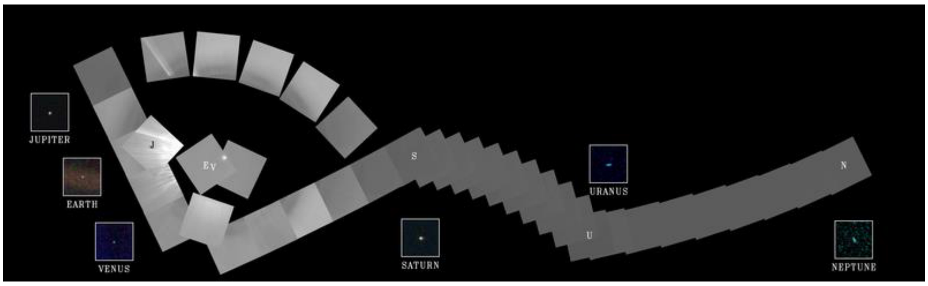
Model Answer

The Saturn montage image displays Jupiter and its moons on the left and Saturn and several of its moons on the right. The text describes Voyager 1's encounter with Saturn, noting that it made a close flyby of Saturn's large moon Titan, which was larger than Earth's Moon and possessed a dense atmosphere, coming within 4,000 miles of its surface near its South Pole. This encounter was significant because it changed Voyager 1's trajectory, preventing it from making any further planetary encounters, which thus began its long journey out of the solar system.

The text does not explicitly state what trade-off made Titan the highest-priority target, nor does it detail why this decision permanently closed Voyager 1's path to the outer planets, or what Voyager 2's different Saturn geometry preserved for the mission as a whole.

Inference: 14.2652 s · E2E: 14.7413 s · CPU: 0.3% · RAM: 0.108 GB · GPU: 26.4% · Images: 1 · FCD: 24.04

Retrieved Images



Family portrait of six planets taken by Voyager 1 on February 14, 1990, when it was 6 B km from Earth. Not exactly close encounters

Query 9 of 14

Question

The Uranus montage and the Neptune-Triton montage each show a planet alongside several moons. Using these images in combination with the text on Voyager 2's encounters, explain why Voyager 2 discovered 11 moons at Uranus and 6 at Neptune despite having far less observing time at each world than it had spent at Jupiter, and what this implies about the relationship between mission geometry and small-body discovery yield.

Retrieved Context

[1] Montage of Uranus and several of its larger moons (left) and of Neptune and its largest moon Triton
(right). Courtesy of NASA.
On February 14, 1990, more than 12 years after it began its journey and shortly before its cameras were
permanently turned off to conserve power, Voyager 1 spun around and pointed them back into the
solar system. In a mosaic of 60 images, it was able to capture a “family portrait” of six of the solar
system’s planets, including a pale blue dot called Earth. It was fitting that these were the last pictures
from either Voyager.
(Source: Voyager Grand Tour PDF.pdf, page 4)

[2] Montage of Jupiter and its four largest moons, (left) and of Saturn and several of its moons (right).
Images are not to scale. Courtesy of NASA.
Two more stops: Uranus and Neptune
Voyager 2’s flyby of Saturn set it up to automatically fly by Uranus after a coast of five and a half years.
It began to study the Uranian system in November 1985 and flew within 50,600 miles of its cloud tops
on January 24, 1986, before concluding its observations in February. In conducting the first ever robotic
reconnaissance of Uranus, Voyager 2 observed the planet’s atmosphere, satellites and thin ring system,
discovering 11 new moons and a previously unknown magnetic field and returning about 8,000 pictures.
And then, it was on to Neptune, arriving there after another coast of three and a half years. Beginning
its study in June 1989, Voyager 2 flew within just 3,076 miles to Neptune’s cloud tops on August 25,
concluding its observations in October. That close pass to Neptune allowed detailed observations of its
largest moon Triton. During this first ever close up study of Neptune, Voyager 2 returned about 10,000
photographs of the planet, its satellites and dark rings, discovering 6 new moons, a Great Dark Spot on
(Source: Voyager Grand Tour PDF.pdf, page 3)

[3] the planet as well as a magnetic field, and geysers on the large moon Triton. After exiting the Neptune
system, Voyager 2 joined its fellow spacecraft on a journey out of the solar system. But the intrepid
Voyagers weren’t quite done yet.
(Source: Voyager Grand Tour PDF.pdf, page 4)

[Image Captions]
[Image 1 Caption] the planet as well as a magnetic field, and geysers on the large moon Triton. After exiting the
Neptune
system, Voyager 2 joined its fellow spacecraft on a journey out of the solar system. But the intrepid
Voyagers weren’t quite done yet. Montage of Uranus and several of its larger moons (left) and of Neptune and its
largest moon Triton
(right). Courtesy of NASA.

Ground Truth Answer

The Uranus montage shows several larger moons around the tilted ice giant, while the Neptune montage shows Triton prominently alongside Neptune. The text states Voyager 2 began studying Uranus in November 1985 and made its closest approach on January 24, 1986, discovering 11 new moons, while the Neptune encounter beginning June 1989 yielded 6 new moons. At Jupiter, which Voyager 2 studied intensively from April to August 1979 — a much longer campaign — only three previously undiscovered small moons were found. The inversion in discovery yield relative to time suggests that proximity geometry and illumination conditions during a fast flyby can be morefavorable for detecting small, faint satellites than a longer but more distant observation campaign. At Uranus andNeptune the extremely close approach distances (50,600 miles and 3,076 miles respectively) provided the angularresolution and lighting angles needed to detect objects that would have remained hidden from farther distances,regardless of integration time.

Model Answer

The provided text and images describe Voyager 2's encounters with the outer solar system, including observations of Uranus and Neptune. Voyager 2 discovered 11 new moons around Uranus and 6 new moons around Neptune during its study of these planets. This discovery occurred after Voyager 2 flew within close proximity to these worlds. For Uranus, Voyager 2 observed the planet's atmosphere, satellites, and thin ring system, discovering 11 new moons and a previously unknown magnetic field. For Neptune, the close pass allowed detailed observations of its largest moon Triton, and Voyager 2 returned about 10,000 photographs of the planet, its satellites, and dark rings, discovering 6 new moons and a Great Dark Spot.

The text notes that Voyager 2's flyby of Uranus occurred after a coast of five and a half years, and its study of Neptune began in June 1989, with a close pass to Neptune on August 25. The text also mentions that Voyager 2 returned about 8,000 pictures of Uranus and about 10,000 photographs of Neptune. While the text does not explicitly state that the observing time at Uranus and Neptune was less than at Jupiter, it does indicate that the close pass to Neptune allowed detailed observations of Triton and the discovery of new moons. This implies that the geometry of the mission, specifically the close flybys, was instrumental in the small-body discovery yield. The close pass to Neptune allowed for detailed observations that led to the discovery of 6 new moons, and the close pass to Neptune also allowed for detailed observations of Triton.

Inference: 28.7903 s · E2E: 29.2248 s · CPU: 0.2% · RAM: 0.109 GB · GPU: 31.5% · Images: 1 · FCD: 14.66

Retrieved Images

the planet as well as a magnetic field, and geysers on the large moon Triton. After exiting the Neptune system, Voyager 2 joined its fellow spacecraft on a journey out of the solar system. But the intrepid Voyagers weren't quite done yet. Montage of Uranus and several of its larger moons (left) and of Neptune and its largest moon Triton (right). Courtesy of NASA.

Query 10 of 14

Question

Curiosity's self-portrait assembled from MAHLI images is shown in the document. Using the image to assess what the composite reveals about rover configuration and condition, and the technical text describing MAHLI's capabilities and arm placement, explain why this imaging method provides more diagnostic value for mission engineers than a single wide-angle shot from a fixed camera would.

Retrieved Context

[1] The mission uses radio relays via Mars orbiters as the
principal means of communication between Curiosity and
the Deep Space Network’s antennas on Earth. In the first
two years after Curiosity’s landing, the orbiters downlinked
48 gigabytes of data from Curiosity.
In April 2004, NASA solicited proposals for specific instru-
ments and investigations to be carried by Mars Science
Laboratory. The agency selected eight of the proposals
later that year and also reached agreements with Russia
and Spain to carry instruments those nations provided.
A suite of instruments named Sample Analysis at Mars
analyzes samples of material collected and delivered by
A self-portrait of Curiosity, built up from pictures taken by its Mars
Hand Lens Imager.
the rover’s arm, plus atmospheric samples. It includes a
gas chromatograph, a mass spectrometer and a tunable
drive, a rocker-bogie suspension system, and cameras
laser spectrometer with combined capabilities to identify a
mounted on a mast to help the mission’s team on Earth
wide range of carbon-containing compounds and deter-
select exploration targets and driving routes. Unlike earlier
mine the ratios of different isotopes of key elements. Iso-
rovers, Curiosity carries equipment to gather and process
tope ratios are clues to understanding the history of Mars’
samples of rocks and soil, distributing them to onboard
atmosphere and water. The principal investigator is Paul
test chambers inside analytical instruments.
Mahaffy of NASA’s Goddard Space Flight Center, Green-
NASA’s Jet Propulsion Laboratory (JPL), Pasadena, Calif.,
builder of the Mars Science Laboratory, engineered Curi-
An X-ray diffraction and fluorescence instrument called
osity to roll over obstacles up to 65 centimeters (25 inch-
CheMin also examines samples gathered by the robotic
es) high and to travel up to about 200 meters (660 feet)
(Source: mars-science-laboratory.pdf, page 3)

[2] samples of rocks and soil, distributing them to onboard
atmosphere and water. The principal investigator is Paul
test chambers inside analytical instruments.
Mahaffy of NASA’s Goddard Space Flight Center, Green-
NASA’s Jet Propulsion Laboratory (JPL), Pasadena, Calif.,
builder of the Mars Science Laboratory, engineered Curi-
An X-ray diffraction and fluorescence instrument called

osity to roll over obstacles up to 65 centimeters (25 inch-
CheMin also examines samples gathered by the robotic
es) high and to travel up to about 200 meters (660 feet)
arm. It is designed to identify and quantify the minerals in
per day on Martian terrain.
rocks and soils, and to measure bulk composition. The
principal investigator is David Blake of NASA's Ames Re-
The rover's electrical power is supplied by a U.S. Depart-
search Center, Moffett Field, Calif.
ment of Energy radioisotope power generator. The multi-
mission radioisotope thermoelectric generator produces
Mounted on the arm, the Mars Hand Lens Imager takes
electricity from the heat of plutonium-238's radioactive
extreme close-up pictures of rocks and soil, revealing
decay. This long-lived power supply gives the mission an
details smaller than the width of a human hair. It can also
operating lifespan on Mars' surface of a full Mars year
focus on hard-to-reach objects more than an arm's length
(687 Earth days) or more. At launch, the generator pro-
away and has taken images assembled into dramatic self-
vided about 110 watts of electrical power to operate the
portraits of Curiosity. The principal investiga- tor is Kenneth
rover's instruments, robotic arm, wheels, computers and
Edgett of Malin Space Science Systems, San Diego.
radio. Warm fluids heated by the generator's excess heat
(Source: mars-science-laboratory.pdf, page 3)

[3] The Mast Camera, mounted at about human-eye height,
Javier Gómez-Elvira of the Center for Astrobiology, Madrid,
images the rover's surroundings in high-resolution stereo
an international partner of the NASA Astrobiology Institute.
and color, with the capability to take and store high-def-
Russia's Federal Space Agency provided the Dynamic
inition video sequences. It ca

[Image Captions]
[Image 1 Caption] A self-portrait of Curiosity, built up from pictures taken by its Mars
Hand Lens Imager. drive, a rocker-bogie suspension system, and cameras mounted on a mast to help the mission's team on Earth select exploration targets and driving routes. Unlike earlier

Ground Truth Answer

The self-portrait shows the full rover body, mast, arm, and wheel configuration in a composite assembled from many close-range images. The text states MAHLI is the Mars Hand Lens Imager mounted on the robotic arm, capable of taking extreme close-up images revealing details smaller than a human hair, and that it can focus on hard-to-reach objects more than an arm's length away. Because the arm repositions the camera at multiple locations and angles, the resulting mosaic provides overlapping, close-range coverage of surfaces that a fixed wide-angle camera would capture only at low resolution and from a single viewpoint. For engineers, this means dust accumulation on solar panel analogs, wheel wear patterns, hardware surface changes, and instrument condition can all be assessed at sub-millimeter detail in routine imaging without requiring dedicated inspection hardware. The method also confirms arm articulation health implicitly — a successful multi-position mosaic is itself evidence of normal arm function.

Model Answer

The self-portrait of Curiosity, built from pictures taken by its Mars Hand Lens Imager, reveals the rover's configuration, which includes the arm, the rocker-bogie suspension system, and cameras mounted on a mast. The text indicates that the Mars Hand Lens Imager is mounted on the rover's arm and provides close-up pictures of rocks and soil, revealing details smaller than the width of a human hair. Furthermore, the text states that the imaging system, which includes an X-ray diffraction and fluorescence instrument, can roll over obstacles up to 65 centimeters high and travel up to about 200 meters. This capability, combined with the images assembled into the self-portrait, provides a detailed view of the rover's physical structure and its interaction with the terrain. In contrast, a single wide-angle shot from a fixed camera would lack this level of detail. The composite images offer a comprehensive view of the rover's physical state and configuration, which is more diagnostic for mission engineers than a single, less detailed view.

Inference: 19.0796 s · E2E: 19.6741 s · CPU: 0.3% · RAM: 0.115 GB · GPU: 28.2% · Images: 1 · FCD: 30.99

Retrieved Images

A self-portrait of Curiosity, built up from pictures taken by its Mars Hand Lens Imager. drive, a rocker-bogie suspension system, and cameras mounted on a mast to help the mission’s team on Earth select exploration targets and driving routes. Unlike earlier

Query 11 of 14

Question

The Cepheid star brightness sequence images from the Andromeda galaxy and the distant supernova comparison images from 1995 and 2002 are presented as two complementary distance-measurement methods. Using both image sets and the text about refining the Hubble constant, explain why astronomers need both techniques rather than relying on only one, and how the combination of refined distances and velocity measurements changed the estimated age of the universe.

Retrieved Context

[1] Above: Astronomers use cyclical changes in the
brightness of Cepheid stars to determine astronomical
distances. The arrow points to a Cepheid star in the
Andromeda galaxy observed by Hubble (inset boxes).
Right: Certain supernovas have a characteristic
maximum brightness that can be used to calculate their
distances from Earth. Refining celestial distances enables
astronomers to better calculate the expansion rate of
the universe. The arrow points to a distant supernova
discovered in an area of the sky first imaged in 1995
called the Hubble Deep Field. Astronomers found the
supernova when they targeted the same area of sky again
in 2002 and saw a change.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 3)

[2] Tracing the Growth of Galaxies
Like documenting a child’s development in a scrapbook,
astronomers have used Hubble to capture the appearance
of many developing galaxies throughout cosmic time. This is
possible because of the mathematical relationship between
cosmic distance and time: the deeper Hubble peers into
space, the farther back it looks in time. As it happens, the
most distant and earliest galaxies spied by Hubble are smaller
and more irregularly shaped than today’s grand spiral and
elliptical galaxies. This is evidence that galaxies grew over
time through mergers with other galaxies to become the giant
systems we see today.
amounts of stars and gas in galaxies, the types and amounts of
identifiable chemical elements present, and star-formation rates.
And the evolution continues. Hubble observations of
our neighboring Andromeda galaxy (M31) have allowed
astronomers to predict with certainty that a titanic collision
between our Milky Way galaxy and Andromeda will inevitably
take place beginning 4 billion years from now. The galaxy is
now 2.5 million light-years away, but it is inescapably moving
toward the Milky Way under the mutual pull of gravity between
the two galaxies and the invisible dark matter that surrounds
them both. The merger will likely result in the creation of a
giant elliptical galaxy billions of years from now.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 4)

[3] Discovering a Runaway Universe

Our cosmos is getting bigger. Nearly a century ago Edwin
Hubble measured the expansion rate of the universe. This
value, called the Hubble constant, is an essential ingredient
needed to determine the age, size, and fate of the cosmos.
Before Hubble was launched, the value of the Hubble
constant was imprecise, and calculations of the universe's age
ranged from 10 billion to 20 billion years. Now, astronomers
using Hubble have refined their estimates of the universe's
present expansion rate and are working to make it more
accurate. They do this by getting better galaxy distance
measurements from Hubble and coupling these values with
the best galaxy-velocity measurements obtained from other
telescopes. Scientists measure distances by comparing the
brightness of a known object in our galaxy (like a star or an
exploded star) to that of a similar object in a distant galaxy.
With Hubble's refined distance values, calculations currently
put the age of the universe as 13.8 billion years.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 3)

[Image Captions]
[Image 1 Caption] Background photo: R. Gendler Above: Astronomers use cyclical changes in the
brightness of Cepheid stars to determine astronomical
distances. The arrow points to a Cepheid star in the
Andromeda galaxy observed by Hubble (inset boxes).

Ground Truth Answer

The Cepheid sequence shows a star in Andromeda varying in brightness across multiple observation dates — this cyclical brightness change is the basis for the period-luminosity distance calibration. The 1995 and 2002 comparison images show a supernova appearing in a previously empty patch of the Hubble Deep Field, illustrating how Type Ia supernovas serve as standard candles at distances where individual Cepheids cannot be resolved. The text explains astronomers measure distances by comparing the known brightness of objects like Cepheid stars or standard supernovas to their apparent brightness in distant galaxies. Cepheids are well-resolved and reliable within the local universe but too faint to detect in very distant galaxies; supernovas are bright enough to see at cosmological distances but need Cepheid-based calibration in the local universe to set their absolute luminosity scale. By combining both ladders, Hubble refined galaxy distance measurements sufficiently to narrow the Hubble constant uncertainty, and coupling those distances with galaxy velocity measurements from other telescopes, calculations currently place the age of the universe at 13.8 billion years, far more precise than the pre-Hubble range of 10 to 20 billion years.

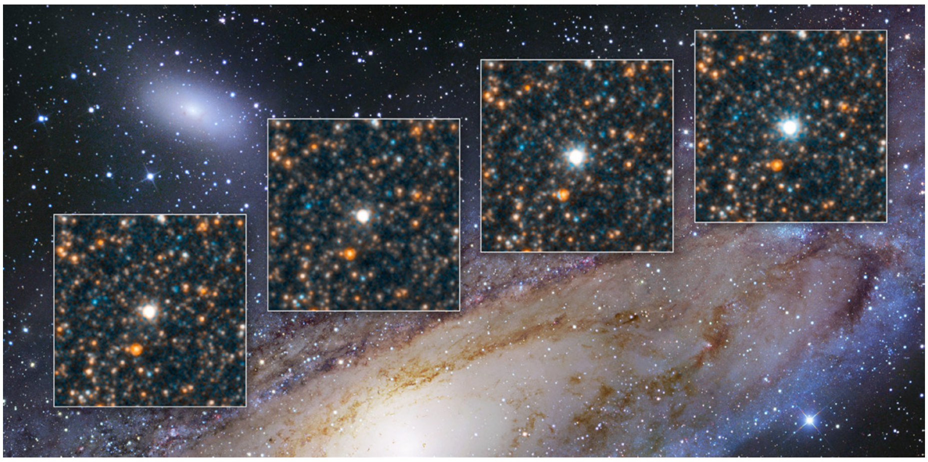
Model Answer

Astronomers use both the Cepheid star brightness sequence images and the distant supernova comparison images as complementary distance-measurement methods to refine their understanding of cosmic distances and the expansion rate of the universe. The text indicates that astronomers use cyclical changes in the brightness of Cepheid stars to determine astronomical distances, and certain supernovae have a characteristic maximum brightness used to calculate their distances from Earth.

Refining celestial distances enables astronomers to better calculate the expansion rate of the universe. Furthermore, the text states that astronomers refine estimates of the universe's present expansion rate by getting better galaxy distance measurements from Hubble and coupling these values with the best galaxy-velocity measurements obtained from other telescopes. This process involves measuring distances by comparing the brightness of a known object to that of a similar object in a distant galaxy. With Hubble's refined distance values, calculations currently put the age of the universe as 13.8 billion years, which is an improvement over previous estimates ranging from 10 billion to 20 billion years.

Inference: 16.4918 s · E2E: 16.9386 s · CPU: 0.3% · RAM: 0.115 GB · GPU: 27.3% · Images: 1 · FCD: 15.63

Retrieved Images



Background photo: R. Gendler Above: Astronomers use cyclical changes in the brightness of Cepheid stars to determine astronomical distances. The arrow points to a Cepheid star in the Andromeda galaxy observed by Hubble (inset boxes).

Query 12 of 14

Question

The M84 galaxy images show a camera view with a dark dust band and a spectrograph plot with dramatic color shifts from blue to red. Using both images and the text on black hole mass measurements, explain exactly what physical phenomenon the color shift encodes, why the spectrograph slit position relative to the core is critical to the measurement, and how this technique established the mass relationship between black holes and their host galaxies.

Retrieved Context

[1] Shown here are two Hubble views of the galaxy M84. Left: This camera view shows the bright core at the center of the galaxy surrounded by
a (vertical) dark band of gas and dust. Right: This plot was generated by passing light near the core of the galaxy through a spectrograph. The
thin, vertical rectangle in the center of the left panel shows the size and shape of the spectrograph's sampling slit. Spectra taken left and right
of the slit's position, which is centered on the core of the galaxy, show a dramatic shift in color to blue or red. Blueshifted light indicates that the
emitting light source is moving toward Earth, while redshifted light indicates objects that are moving away. At 880,000 miles per hour, stars and
glowing gases nearest to the core of M84 are moving the fastest. They are circling a black hole at the center of the galaxy, and so appear to be
moving rapidly toward Earth on the left and receding on the right.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 7)

[2] Top left: This sketch shows paths of light from a distant galaxy that is being gravitationally lensed. Top right: Gravitational lensing has created
three distorted images of the same background galaxy and five of a background quasar in this image of galaxy cluster SDSS J1004+4112.
Bottom: These two pictures show a massive galaxy cluster, Cl 0024+17. The visible-light image (left) shows blue arcs among the yellowish
galaxies. These arcs are magnified, warped images of galaxies located behind the cluster whose light has been distorted and bent. The blue
overlay (right) shows the dark matter density needed to account for the gravitational distortions.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 6)

[3] Shining a Light on Dark Matter
Dark matter is an invisible form of matter that makes up most of the universe's mass and creates its underlying structure. Dark
matter's gravity drives normal matter (gas and dust) to collect and build up into stars and galaxies. Although astronomers cannot
see dark matter, they can detect its influence by observing how the gravity of massive galaxy clusters, which contain dark matter,
bends and distorts the light of more distant galaxies located behind the cluster. This phenomenon is called gravitational lensing.
Hubble's uniquely sharp vision allows astronomers to map the distribution of dark matter in space using gravitational lensing.
Large galaxy clusters contain both dark matter and normal matter. By observing the areas around massive clusters of galaxies
astronomers can identify warped background galaxies and reverse-engineer their distortions to reveal where the densest
concentrations of matter lie. Mathematical models of these results shed light on the location and properties of the lensing

material, both visible and invisible (dark). The universe appears to have about five times more dark matter than regular matter
and seems to be organized around an immense network of dark matter filaments that have grown over time. Massive visible
structures, like galaxy clusters, are found at the intersections of these filaments.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 6)

[Image Captions]
[Image 1 Caption] Left: Hubble’s Ultra Deep Field is one of the most distant looks into
space ever. The cumulative exposure time needed to capture the
image was about a million seconds (11 days). Right: A sample of the faintest and farthest galaxies in the Hubble Ultra Deep Field shows that they were irregularly shaped and
frequently interacting in the distant past.

Ground Truth Answer

The camera image shows M84's bright core crossed by a dark vertical band of gas and dust, while the spectrograph plot shows a sharp transition from blue on one side of the core to red on the other. The text explains that blueshifted light indicates the emitting source is moving toward Earth while redshifted light indicates recession — stars and gas nearest the core are orbiting a central black hole at 880,000 miles per hour, so material on one side of the rotation axis approaches Earth while material on the other side recedes. The spectrograph slit must be positioned centered on the core because the Doppler shift gradient is steepest where the orbital velocities are highest, right at the innermost region around the black hole. The width and sharpness of the blue-to-red transition encodes the rotational velocity, which through Kepler's laws yields the enclosed mass. Applying this technique across many galaxies in a Hubble census revealed that black hole mass scales with the mass of the host galaxy's central stellar bulge — the larger the galaxy, the larger the black hole — suggesting black holes and their galaxies co-evolved.

Model Answer

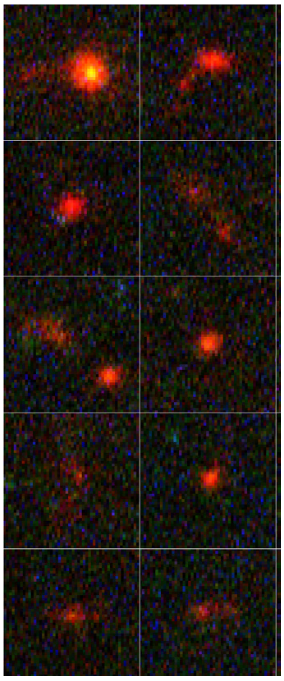
The images of the M84 galaxy show a camera view with a bright core surrounded by a vertical dark band of gas and dust, and a spectrograph plot showing a dramatic shift in color to blue or red on either side of the slit position centered on the core. This color shift encodes the Doppler effect, where blueshifted light indicates that the emitting light source is moving toward Earth, and redshifted light indicates objects that are moving away.

The spectrograph slit position relative to the core is critical because the light from the regions near the core of the galaxy is analyzed. The text indicates that stars and glowing gases nearest to the core of M84 are moving at the fastest speeds, circling a black hole at the center of the galaxy, and thus appear to be moving rapidly toward Earth on the left and receding on the right.

This technique established the mass relationship between black holes and their host galaxies by observing these velocity shifts. The rapid motion of the gas and stars near the core, which is attributed to the gravity of the black hole, allows astronomers to measure these velocities, which in turn relates to the mass of the central object.

Inference: 19.1629 s · E2E: 19.6043 s · CPU: 0.2% · RAM: 0.125 GB · GPU: 29.7% · Images: 1 · FCD: 13.78

Retrieved Images



Left: Hubble's Ultra Deep Field is one of the most distant looks into space ever. The cumulative exposure time needed to capture the image was about a million seconds (11 days). Right: A sample of the faintest and farthest galaxies in the Hubble Ultra Deep Field shows that they were irregularly shaped and frequently interacting in the distant past.

Query 13 of 14

Question

The Jupiter impact sequence showing cloud darkening from Comet Shoemaker-Levy 9, the close-up plume image from the first large fragment impact, and the Saturn aurora image are all on the same or adjacent pages. Using these three images and the text covering both the impact events and aurora observations, explain what each image type contributes uniquely to understanding planetary atmospheric dynamics, and why continuous ultraviolet monitoring capability is the common prerequisite for all three scientific results.

Retrieved Context

[1] This image series, starting in the lower right corner, shows the darkening of Jupiter’s clouds as a result of the impact of Comet Shoemaker-Levy 9.
Hubble captured a colossal plume rising above the limb of Jupiter from the impact of the first large fragment of Comet Shoemaker-Levy 9. (Times along the left side are in hour:minute format.)
Jupiter’s moons also have yielded important clues in the search for life beyond Earth. Hubble provided the best evidence yet for an underground saltwater ocean on Ganymede, the largest moon in the solar system, by detecting related activity in Ganymede’s own auroras. This subterranean ocean is thought to have more water than all the water on Earth’s surface.
Hubble also recorded evidence of transient changes in the atmosphere above the surface of Jupiter’s moon Europa. Astronomers suspect that these disturbances are caused by gas plumes expelled from a subsurface ocean. Identifying liquid water is crucial in the search for habitable worlds beyond Earth and in the quest to find life as we know it.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 8)

[2] Studying the Outer Planets and Moons
Hubble has witnessed impacts on Jupiter that were produced by minor bodies in the solar system. The latest collision observed by Hubble occurred in 2009 when a suspected asteroid plunged into Jupiter’s atmosphere, leaving a temporary dark feature the size of the Pacific Ocean. In 1994 Hubble watched 21 fragments of Comet Shoemaker-Levy 9 bombard the giant planet sequentially—the first time astronomers witnessed such an event. Each impact left a temporary black, sooty scar within Jupiter’s clouds.
Jupiter is well known for its Great Red Spot, a giant storm roughly the size of Earth that has been continuously visible since at least the late 1800s. The mammoth storm has been shrinking in size for at least 80 years. Astronomers now use Hubble regularly to measure the Red Spot’s size and investigate why it is slowly disappearing.
Hubble also made the first images of bright auroras at the northern and southern poles of Saturn and Jupiter. Auroras are brilliant curtains of light that appear in the upper atmosphere of a planet with a magnetic field. They develop when electrically charged particles trapped in the magnetic field spiral inward at high speeds toward the north and

south magnetic poles. When these particles hit the upper
atmosphere, they excite atoms and molecules there causing
them to glow in a similar process to that of a fluorescent light.
This image series, starting in the lower right corner, shows the
darkening of Jupiter’s clouds as a result of the impact of Comet
Shoemaker-Levy 9.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 8)

[3] Top left: Jupiter’s trademark Great Red Spot—a swirling, 300-mile-per-hour storm—has shrunk to its smallest size ever. Astronomers have
followed this gradual downsizing since the 1930s and are now keenly monitoring its shrinkage and investigating its cause with Hubble. Top
right: Hubble’s ability to capture ultraviolet light revealed a region of glowing auroras over the pole of Saturn. While sharing some features of
Earth’s auroras and others of Jupiter’s, Saturn’s auroras are different from both and are likely unique in the solar system. Below left: The blue
areas mapped on Jupiter’s moon Europa mark spots where Hubble found spectroscopic evidence of oxygen and hydrogen—the two elements
that form the water molecule. Scientists believe this is most likely the result of water vapor plumes erupting from beneath the surface, with
recent visual evidence of these plumes also seen by Hubble (below right). The pictures of Europa are a combination of shots gathered by
NASA’s Voyager and Galileo missions.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 9)

[Image Captions]
[Image 1 Caption] Studying the Outer Planets and Moons This image series, starting in the lower right corner, shows the
darkening of Jupiter’s clouds as a result of the impact of Comet
Shoemaker-Levy 9. Hubble captured a colossal plume rising above the limb of Jupiter
from the impact of the first large fragment of Comet Shoemaker-Levy 9.
(Times along the left side are in hour:minute format.)

Ground Truth Answer

The Shoemaker-Levy 9 cloud darkening sequence shows Jupiter's atmosphere progressively accumulating impact scars as 21 fragments struck sequentially in 1994 — each dark sooty scar documents energy deposition depth and lateral spread in the Jovian cloud deck. The plume image from the first large fragment shows a colossal debris column rising above Jupiter's limb in time-stamped frames at minute-level cadence, capturing the transient ejecta dynamics during the first seconds to minutes after impact. The Saturn aurora image shows UV-bright curtains of light at Saturn's polar regions, capturing magnetospheric particle precipitation into the upper atmosphere. Each contributes something distinct: the scar sequence measures integrated impact effects over days to weeks; the plume sequence captures real-time ballistic ejecta dynamics; the aurora image characterizes steady-state magnetospheric coupling. All three depend on ultraviolet sensitivity because: impact plumes and atmospheric ejecta are bright in UV; auroral emission from excited atmospheric molecules peaks in the UV; and the temporal fidelity needed to capture transients requires space-based UV capability free from atmospheric UV absorption that blocks ground-based observation.

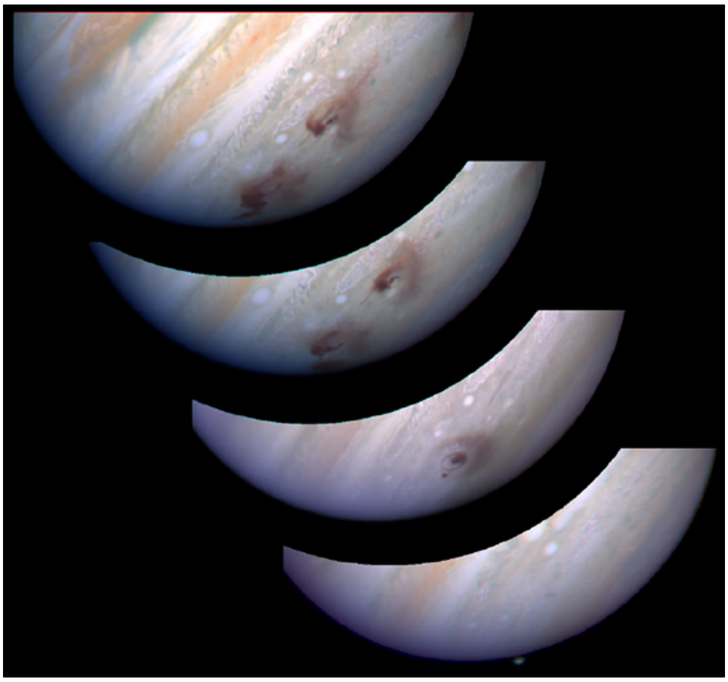
Model Answer

The image series showing the darkening of Jupiter’s clouds due to the impact of Comet Shoemaker-Levy 9, the close-up plume image from the first large fragment impact, and the Saturn aurora image each contribute uniquely to understanding planetary atmospheric dynamics. The image series of Jupiter's cloud darkening demonstrates the direct physical effect of the impact on the atmosphere, as noted by Hubble capturing a colossal plume rising above the limb of Jupiter from the impact of the first large fragment of Comet Shoemaker-Levy 9. This relates to the text which mentions the impact events. The Saturn aurora image reveals the phenomenon of auroras over the pole of Saturn, which are caused by electrically charged particles interacting with the upper atmosphere due to a magnetic field.

Continuous ultraviolet monitoring capability is the common prerequisite for all three scientific results because the text indicates that Hubble's ability to capture ultraviolet light was used to reveal the glowing auroras over Saturn. Furthermore, the text discusses the detection of transient changes in the atmosphere above Europa caused by gas plumes, suggesting that observing atmospheric changes and phenomena like auroras requires sensitivity to light that reveals these interactions, which is facilitated by ultraviolet observation.

Inference: 19.9611 s · E2E: 20.4353 s · CPU: 0.2% · RAM: 0.127 GB · GPU: 30.4% · Images: 1 · FCD: 18.53

Retrieved Images



Studying the Outer Planets and Moons This image series, starting in the lower right corner, shows the darkening of Jupiter’s clouds as a result of the impact of Comet Shoemaker-Levy 9. Hubble captured a colossal plume rising above the limb of Jupiter from the impact of the first large fragment of Comet Shoemaker-Levy 9. (Times along the left side are in hour:minute format.)

Query 14 of 14

Question

The TW Hydrae disk image shows a gap structure around the star, and the 1997 and 2012 Beta Pictoris disk images show an edge-on disk changing over fifteen years. Using both sets of images alongside the planet-formation text, explain what specific morphological features in each image constitute evidence for an embedded or nearby planet, why temporal baseline matters for the Beta Pictoris case but not for TW Hydrae, and how these two systems together support a dynamic rather than static picture of disk evolution.

Retrieved Context

[1] This Hubble image and illustration show a gap in a protoplanetary disk of dust and gas around the nearby star TW Hydrae. The gap's presence
is probably due to the effects of a growing, unseen planet that is gravitationally sweeping up material and carving out a lane in the disk.
Astronomers used a masking device within the Hubble camera to block out the star's bright light so that the disk's structure could be seen.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 17)

[2] Finding Planetary Construction Zones
Astronomers used Hubble to confirm that planets form in dust
disks around stars. The telescope first resolved protoplanetary
disks around nearly 200 stars in the bright Orion Nebula.
Looking at nearby stars elsewhere in the sky, Hubble
completed the largest and most sensitive visible-light imaging
survey of dusty debris disks, which were probably created by
collisions between leftover objects from planet formation.
Two particular stars illustrate these findings: TW Hydrae
and Beta Pictoris. Using a mask to block the star's bright
light, Hubble scientists spotted a mysterious gap in a vast
protoplanetary disk of gas and dust swirling around the star TW
Hydrae. The gap is most likely caused by a growing, unseen
planet that is gravitationally sweeping up material and carving
out a lane in the disk like a snowplow. It is 1.9 billion miles wide
and not yet completely cleared of material. Researchers have
also noted changes in the planetary disk surrounding Beta
Pictoris. By masking out the star's light, scientists have studied
changes in the orbiting material caused by a massive planet
embedded within its dust disk. Astronomers spotted the planet
using the European Southern Observatory.
Here are a few of the many
protoplanetary disks that were
discovered by Hubble in the
large, majestic Orion Nebula.
The disks appear as silhouettes,
illuminated from behind by the
glowing gases of the nebula.
Below left and right: Hubble has provided detailed pictures showing
changes in the large, edge-on, gas-and-dust disk encircling the
20-million-year-old star Beta Pictoris.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 17)

[3] Here are a few of the many

protoplanetary disks that were
discovered by Hubble in the
large, majestic Orion Nebula.
The disks appear as silhouettes,
illuminated from behind by the
glowing gases of the nebula.
Below left and right: Hubble has provided detailed pictures showing
changes in the large, edge-on, gas-and-dust disk encircling the
20-million-year-old star Beta Pictoris.
(Source: highlights_of_hubbles_exploration_of_the_universe.pdf, page 17)

[Image Captions]
[Image 1 Caption] Below left and right: Hubble has provided detailed pictures showing
changes in the large, edge-on, gas-and-dust disk encircling the
20-million-year-old star Beta Pictoris.

Ground Truth Answer

The TW Hydrae image shows a protoplanetary disk with a gap approximately 1.9 billion miles wide that is not yet completely cleared of material. The text explains this gap is most likely caused by a growing, unseen planet gravitationally sweeping up material like a snowplow, carving a lane in the disk. The morphological evidence is the gap itself — its width, incomplete clearing, and position within the disk are all consistent with an embedded planet at that orbital radius. The Beta Pictoris pair shows the large edge-on disk in 1997 and 2012; the text states scientists studied changes in the orbiting material caused by a massive planet embedded within the dust disk. For TW Hydrae, a single image suffices because the gap is a static structural feature that persists on timescales far longer than an observation campaign. For Beta Pictoris, the 15-year baseline is essential because the planet-induced changes in disk structure — warps, asymmetries, or material redistribution — are only detectable as differences between epochs, not from any single snapshot. Together, TW Hydrae shows a planet actively carving its initial gap while Beta Pictoris shows a more evolved system where a mature embedded planet continues to dynamically reshape disk material over decadal timescales.

Model Answer

The image of the TW Hydrae disk shows a gap in the protoplanetary disk of dust and gas around the star, which is probably due to the effects of a growing, unseen planet gravitationally sweeping up material and carving out a lane in the disk. This gap is evidence for an embedded planet.

In contrast, the images of the Beta Pictoris disk, which are edge-on views, show changes in the large, gas-and-dust disk over time. This temporal baseline matters for the Beta Pictoris case because scientists have studied changes in the orbiting material caused by a massive planet embedded within its dust disk.

Together, these two systems support a dynamic rather than static picture of disk evolution. The TW Hydrae image shows a gap indicative of a planet actively carving the disk, while the Beta Pictoris images illustrate changes in the disk material over time, demonstrating that protoplanetary disks are evolving due to the presence of embedded planets.

Inference: 15.1540 s · E2E: 15.5213 s · CPU: 0.2% · RAM: 0.128 GB · GPU: 26.3% · Images: 1 · FCD: 12.68

Retrieved Images

Below left and right: Hubble has provided detailed pictures showing changes in the large, edge-on, gas-and-dust disk encircling the 20-million-year-old star Beta Pictoris.